



AFRICA
ENERGY FINANCE
Business & Financial Models

CASE 2

CASE 2: SolaraPay Ltd

Financing a PAYGo solar company. Worked case for MODEL 2, version 0.8 (development build), and Book 2, PAYGo Solar Finance

AUTHOR & IDEATION

Emmanuel Boujiekka Kamga

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Case summary

Teaching case. SolaraPay Ltd and the Republic of Kivara are fictional. Every figure, including the 24 month portfolio history, is synthetic teaching data built under a fixed random seed so that the case can be reproduced exactly. All numbers quoted here come from the case workbook `SolaraPay_Case_Model_v0.8-dev.xlsx`, a version of MODEL 2 (version 0.8, development build) loaded with that history. The projections are unchanged from version 0.7, whose outputs were reproduced to rounding precision by an independent recalculation of the full model. Book 2, Chapter 16 walks through the same case.

The decision. SolaraPay Ltd, a PAYGo solar distributor in the Republic of Kivara (currency: the Kivara shilling, KVS; opening rate KVS 135 per USD), is raising USD 4.0 million of equity from an impact fund at a pre money valuation of USD 8.0 million, which gives the fund a 33.3% stake. In parallel it has asked a local bank for a KVS 4.5 billion receivables facility. The fund's investment committee must choose between Go, Conditional Go and Stop.

Recommendation: Conditional Go. The business can create value, but the management plan overstates repayment, recoveries and covenant headroom. The investment holds up only if conditions are attached on pricing, Tier 4 and 5 exposure, collections, recovery evidence and valuation (Section 9). The workbook's rule reads STOP on a failed test: the calibrated Base fails the lender's annual DSCR test in Years 1 to 4, a finding against the business plan as modelled, and most critical readiness gates are not yet evidenced. The Conditional Go is the analyst's judgement, explicitly conditional on the DSCR covenant being replaced or reset with the lender and on the critical gates the conditions address being evidenced before disbursement.

Key figure (calibrated Base unless stated)	Value
Revenue, Year 1 and Year 5	USD 8.1m and USD 51.2m
EBITDA margin, Year 5	16.4% (management plan 18.6%)
Peak equity required	USD 9.0m; no top up needed in Base
Investor IRR and multiple (USD)	29.0% and 3.6x (management plan 33.1% and 4.2x)
Investor outcome, Downside	Total loss (exit equity of zero)
Lowest trailing three month operational collection rate against a 70% covenant	70.0%, with no headroom
Observed operational collection rate, last twelve months of history	69.3%, already below the proposed covenant
Investment readiness (workbook)	4 of 23 gates met (4 of 13 critical); decision rule reads STOP on a failed test (annual DSCR below 1.20x in Years 1 to 4)

1. Company and market

SolaraPay has sold solar products on pay as you go terms in Kivara for two years. Customers pay a deposit and then a daily rate through mobile money; the device locks remotely when payments stop and unlocks permanently at the end of the plan.

The company sells Tiers 1 to 3 today. Its plan adds two solar inverter systems with lithium storage (Tiers 4 and 5), aimed at urban households and small businesses on an unreliable grid.

Tier	Product	Cash price (KVS)	Implied APR	Tenor	Planned sales mix
1	Pico solar kit, 10 Wp	6,500	106%	12 months	25%
2	SHS with TV, 80 Wp	39,000	50%	24 months	40%
3	Large SHS with DC fridge, 300 Wp	117,000	39%	30 months	22%
4	Inverter with lithium, about 1.2 kWp and 5 kWh (new)	390,000	32%	36 months	9%
5	Inverter with lithium, about 3 kWp and 10 kWh (new)	845,000	25%	48 months	4%

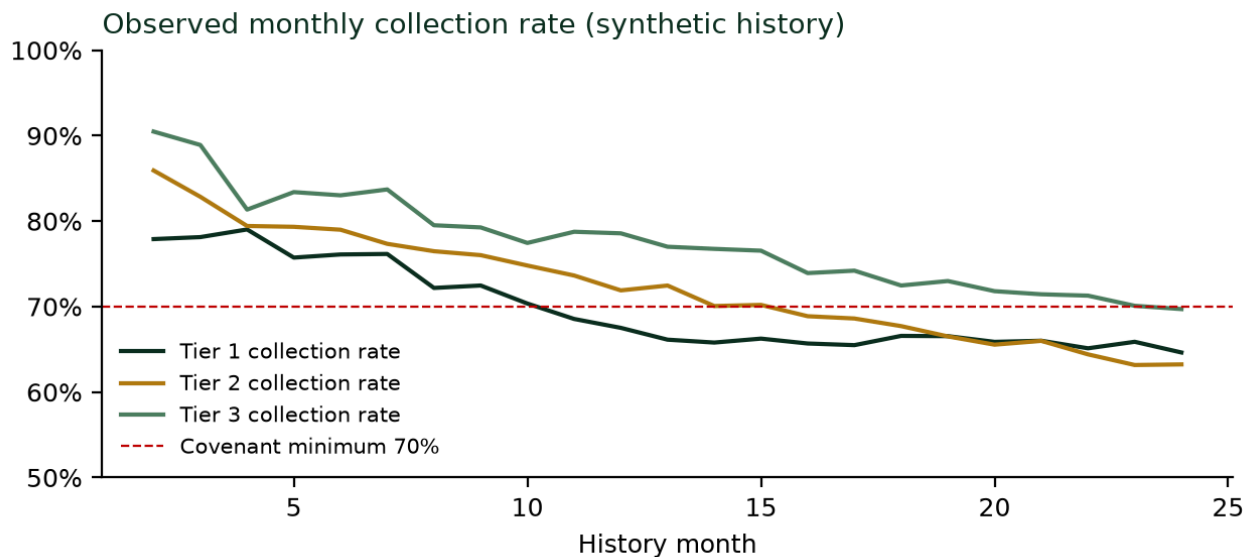
Year	1	2	3	4	5
Units sold	9,000	18,000	28,000	36,000	42,000

The proposed capital structure has three layers. Initial equity totals USD 9.0m (KVS 1,215m), including the fund's USD 4.0m. A USD 3.0m term loan carries 10% interest, a twelve month grace period and 36 months of amortisation. The KVS 4.5bn receivables facility is priced at 16% and advances 50% of eligible Tier 1 receivables, 70% for Tiers 2 and 3 and 75% for Tiers 4 and 5, drawn as needed.

2. Testing the plan against SolaraPay's own data

Management built its plan on the model's default credit assumptions: a monthly default hazard of 3.5% for Tier 1, 2.6% for Tier 2 and 1.9% for Tier 3, and collection rates on paying accounts of 88% to 90%.

Before projecting anything, the analyst loaded SolaraPay's 24 month portfolio history into *Credit_Input* and eighteen monthly cohorts per tier into *Vintage_Input*, and switched the credit data mode to Actual.



The history tells a consistent story.

Collections are weakening as the book seasons. The portfolio collection rate fell from 77.4% in the first year of history to 69.3% in the second. In the latest month, *Credit_Portfolio* reports a collection ratio of 66.5%, 17.0% of receivables more than 30 days past due and indicative ECL coverage of 36.5%.

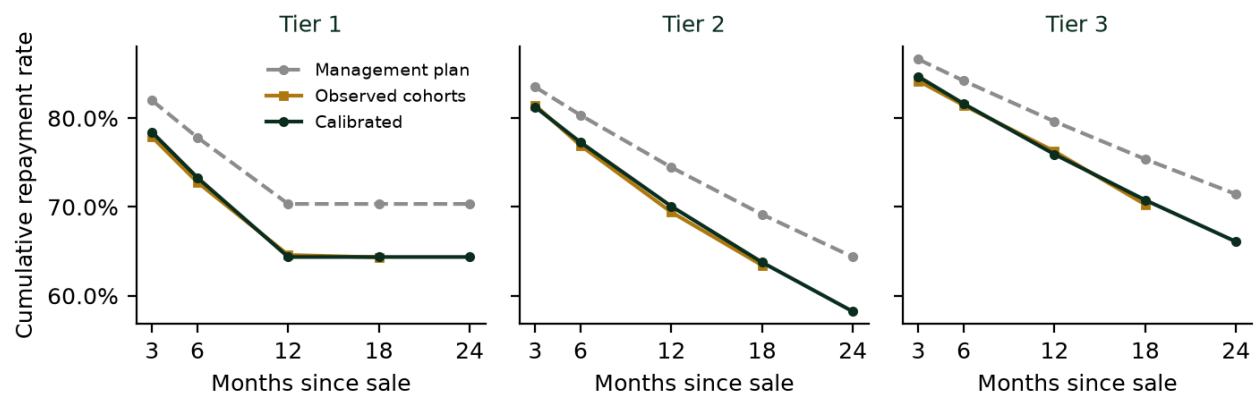
Cohorts repay below plan at every age. At month 12, observed cumulative repayment compares with the plan as follows:

Tier	Observed	Plan
Tier 1	64.6%	70.3%
Tier 2	69.4%	74.5%
Tier 3	76.3%	79.6%

Recoveries are close to nil. The observed LGD proxy is about 99%, meaning almost nothing comes back from repossessed units, whereas the plan assumes that 20% to 70% of defaulted units are repossessed, depending on the tier.

There is no ownership evidence yet. No cohort is old enough to measure ownership at twice the tenor, so an ownership linked RBF programme cannot pay for now, and the model correctly pays zero under that design.

Vintage repayment curves: plan, observed cohorts and calibration



Recalibration. The analyst re-estimated Tier 1 to 3 credit behaviour from the cohort data. The calibrated curves (green) sit on the observed points (gold).

Tier	Monthly default hazard, plan	Calibrated	Collection on paying accounts, plan	Calibrated
1	3.50%	4.55%	88%	86%
2	2.60%	3.38%	88%	87%
3	1.90%	2.47%	90%	89%
4 and 5	Unchanged, no history	Unchanged	Unchanged, no history	Unchanged

Lesson. The most productive hour of this analysis went into SolaraPay's own cohort data, not into the projections. A plan built on assumptions that the company's history already contradicts cannot serve as a base case.

3. Unit economics by tier

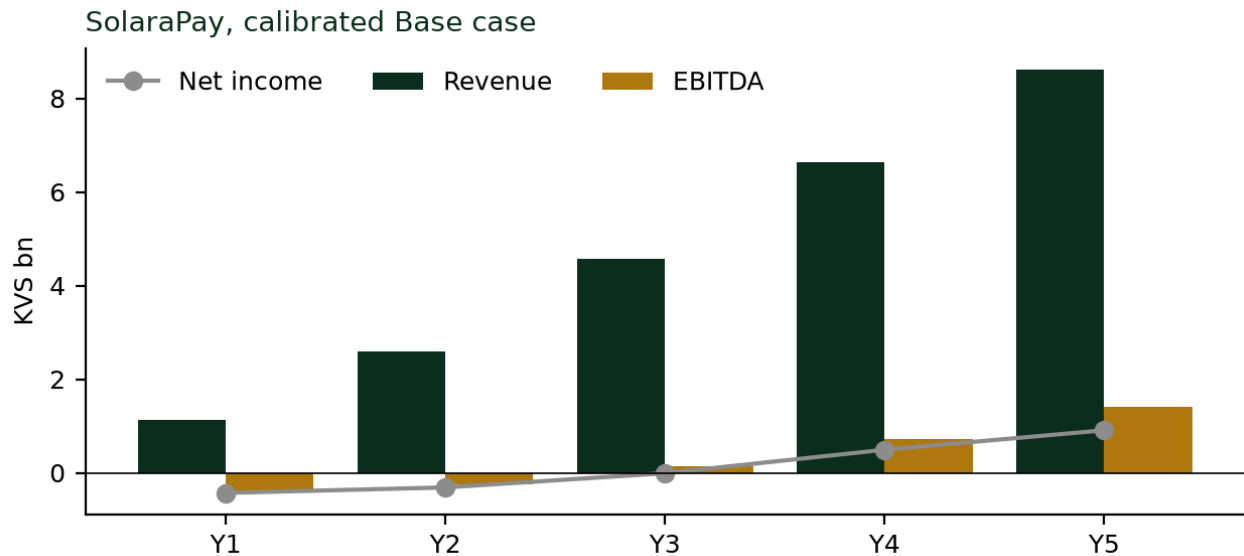
Calibrated Base, from *Unit_Economics*:

Tier	Expected loss rate	Lifetime contribution (KVS)	LTV to CAC	Cash payback	Unit IRR
1	35.6%	1,092	2.4x	9 months	66%
2	41.8%	7,524	3.5x	17 months	37%
3	38.2%	33,897	5.5x	18 months	47%
4	27.1%	180,443	11.0x	17 months	67%
5	26.6%	416,975	14.9x	22 months	54%

Every tier still creates value per unit, but the distribution matters. Tier 2, which carries 40% of planned volume, is the weakest product per unit: it loses 41.8% of scheduled instalments and takes 17 months to pay back. Tiers 4 and 5 look the most attractive, with LTV to CAC of 11x to 15x, yet their credit assumptions have no history behind them.

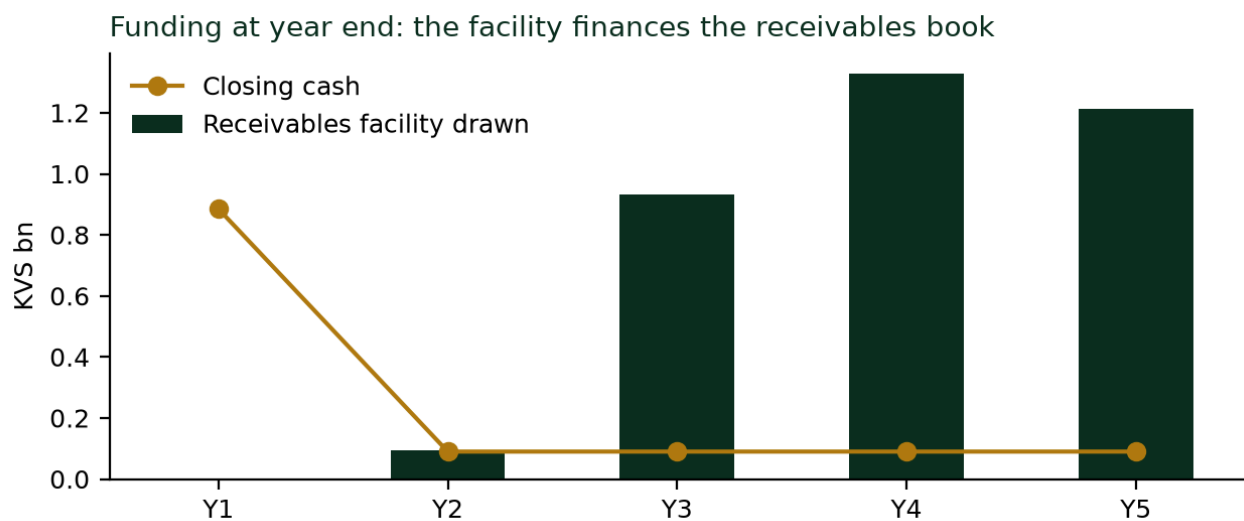
Affordability. On *Consumer_Risk*, the instalment exceeds the model's 10% payment burden threshold (a model policy threshold, not a standard) for three tiers: Tier 2 at 13.0% of illustrative household income, Tier 3 at 13.2% and Tier 4 at 11.3%. A high burden is a leading indicator of the defaults already visible in Tier 2. Tier 1's implied APR of 106% will draw regulatory and reputational attention.

4. Calibrated projections

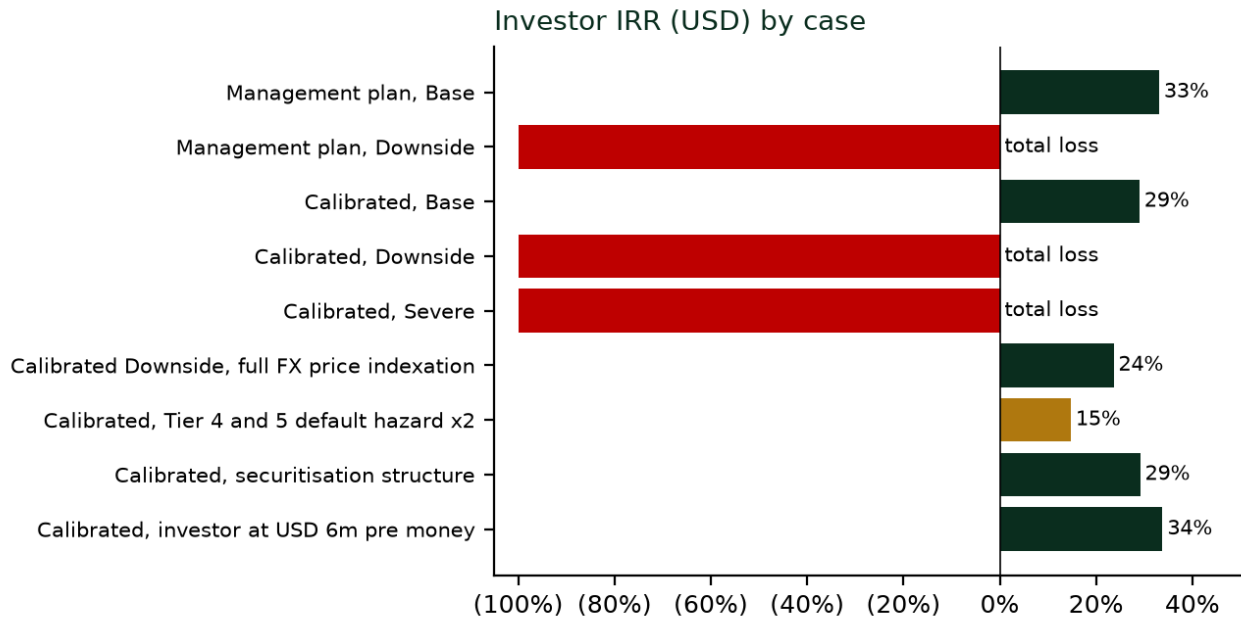


Calibrated Base	Y1	Y2	Y3	Y4	Y5
Revenue (KVS bn)	1.13	2.60	4.58	6.63	8.62
Gross margin	41.5%	46.8%	50.7%	54.0%	56.6%
EBITDA margin	(35.2%)	(9.4%)	2.9%	10.9%	16.4%
Net income (KVS bn)	(0.43)	(0.31)	(0.01)	0.49	0.91
Net credit losses to revenue	37.5%	33.9%	31.3%	29.2%	27.5%
Operational collection rate	82.3%	76.8%	73.3%	71.5%	70.3%
30+ DPD to gross receivables	8.4%	11.4%	12.8%	13.7%	14.4%
Debt to book equity (year end)	0.54x	0.98x	2.51x	1.47x	0.65x

EBITDA turns positive in month 29, and operating cash flow stays positive from month 51 to the end of the horizon. The USD 9.0m of initial equity is sufficient in Base: the receivables facility finances the growing book and peaks at KVS 1.39bn in month 54. The annual DSCR stays below 1.20x in Years 1 to 4. A cash flow DSCR covenant is ill suited to a PAYGo book in growth; portfolio covenants are the better instrument (Section 7).



5. Stress tests



Case	Peak equity (USD m)	Y5 EBITDA margin	Y5 operational collection rate	Investor IRR	Multiple	Months in covenant breach
Management plan, Base	9.0	18.6%	73.0%	33.1%	4.2x	0
Management plan, Downside	9.0	3.7%	66.8%	Total loss	0.0x	39
Calibrated, Base	9.0	16.4%	70.3%	29.0%	3.6x	0
Calibrated, Downside	9.0	1.3%	63.9%	Total loss	0.0x	40
Calibrated, Severe	41.9	(25.4%)	55.4%	Total loss	0.0x	45
Calibrated Downside with full FX price indexation	9.0	14.2%	64.4%	23.6%	2.9x	38
Calibrated, Tier 4 and 5 default hazard doubled	9.0	10.3%	63.8%	14.8%	2.0x	38
Calibrated, securitisation structure	9.0	16.4%	70.3%	29.1%	3.6x	0
Calibrated, investor entering at USD 6m pre money	9.0	16.4%	70.3%	33.8%	4.3x	0

Four conclusions follow.

The Downside wipes out the equity. At the Year 5 exit, net debt exceeds six times a much reduced EBITDA, so exit equity is zero. No new equity is called during the five years, but the stake is worth nothing at exit.

Pricing power is the strongest mitigant available. Indexing new contract prices fully to the exchange rate turns the Downside from a total loss into a 23.6% IRR.

Tier 4 and 5 credit is the largest unproven risk. Doubling the default hazard on those two tiers halves the IRR to 14.8% and produces 38 months of covenant breaches.

Structure and entry price matter less than credit, though they are not negligible. A securitisation structure barely moves returns, while entering at a USD 6m pre money valuation lifts the IRR to 33.8%.

6. Valuation and returns

Metric (calibrated Base)	Value
DCF enterprise value at entry (22% discount rate)	USD 6.1m
Exit equity value, 100%, end of Year 5 (6x EBITDA less net debt)	USD 42.9m
Investor stake	33.3%
Investor IRR and multiple (USD)	29.0% and 3.6x

Why the DCF is low. The DCF value sits below the USD 8m pre money valuation, and the terminal value accounts for more than the whole enterprise value. Both are normal for a PAYGo company in its growth phase. Each sale on credit is an investment in working capital, so free cash flow over the first five years is negative in aggregate and the value is concentrated in the terminal year.

The fund should therefore treat the entry price as a position on the exit multiple and on credit quality, and negotiate protection accordingly: a lower pre money valuation, a ratchet, or staged tranches tied to cohort performance.

7. Lender view

Test (calibrated Base, projection)	Result
Trailing three month collection rate, minimum 70%	Lowest 70.0%, no headroom
Receivables at risk, maximum 15%	Highest 14.4%
30+ DPD maximum 25%; 90+ DPD maximum 18%	Within limits
Debt to book equity, maximum 3.0x	Highest at year end 2.51x (Year 3)
Annual DSCR, minimum 1.20x	Below the minimum in Years 1 to 4
Borrowing base (eligibility up to 30 DPD, advance rates of 50% to 75% by tier)	Facility always within the base

The projection passes the monthly covenants, but only just, and fails the annual DSCR test in Years 1 to 4. On actual data the position is weaker: the latest monthly collection ratio of 66.5% is already below the 70% covenant level, and the last twelve months of history averaged 69.3%, so a 70% trailing covenant would very likely be in breach from the first test date.

A lender has two sound options. The first is to set the collection covenant at a level the book can meet, for example 65% with a cure period and a collections improvement plan. The second is to defer the facility until the collection trend has turned. In either case the bank should replace the DSCR test with portfolio covenants (the case workbook keeps the test as drafted: *Covenant definition on Inputs* is set to 1; once the facility has no DSCR test, set it to 0) and require monthly reporting of operational and lender metrics: the operational collection rate, receivables at risk, the write off ratio and cohort repayment curves. These are not PAYGo PERFORM 2026 KPIs. The five PERFORM 2026 KPIs (GOGLA Technical Guide, June 2026) can be computed only from company contract data entered on *PERFORM_2026*, which SolaraPay has not yet provided; the bank should ask for that data too. Any request to extend eligibility beyond 30 DPD should be refused.

8. Benchmarks and calibration

Benchmark_Compare and *Calibration* set the calibrated Base against the workbook's external references, and none can be used as a test. The scale references for net margin, credit losses and return on equity rested on M-KOPA group figures that conflict between sources, and the only primary filing held concerns M-KOPA UK LIMITED, a subsidiary, rather than the group. The M-KOPA comparison used in earlier drafts of this case is therefore withdrawn until the group's consolidated accounts are read. The sector collection rate is reported in secondary sources, but the primary report has not yet been read. *Calibration* shows every reference as suspended.

Ratio, Year 5	SolaraPay	Reference
Net margin	10.5%	None verified (comparison withdrawn)
Expected credit losses to financing revenue	1.28x	None verified
Return on equity	49.8%	None verified
Operational collection rate (not a PERFORM KPI)	70.3%	ESMAP Off-Grid Solar Market Trends Report 2024, reported to give an average PAYGo collection rate of about 62% for 2021 to 2023 (context only; primary document pending)

The ratios still speak for themselves. Expected credit losses at 1.28 times financing revenue mean that financing income does not cover the losses it is meant to price; the hardware margin is carrying them. A net margin of 10.5% and a return on equity of 49.8% in Year 5 sit uneasily with a history of falling collections, and the cost and credit assumptions need to be justified on the company's data rather than against a peer. The sector figure is a collection rate, not a PERFORM repayment rate, and it is not yet verified. The benchmark that matters is SolaraPay's own cohort history, set against the plan in *Vintage_Dashboard*.

9. Recommendation and conditions

Conditional Go: invest USD 4.0m, subject to seven conditions.

1. **Pricing policy.** Contractual or board approved indexation of new contract prices to the KVS/USD rate, which corresponds to setting pass through to 100% in the model. This is the most effective Downside mitigant identified.
2. **Tier 4 and 5 pilot cap.** Tiers 4 and 5 are held to their planned 13% of units and released in tranches only after twelve months of cohort data confirm repayment within 10% of plan, with portfolio at risk reported separately for these tiers.
3. **Collections plan before the facility.** A written collections improvement plan with monthly targets, and a facility covenant set at a level the book can meet (65% trailing collection with a cure period), stepping up as performance improves. The annual DSCR covenant, which the projection fails in Years 1 to 4, is replaced by portfolio covenants or reset with the lender (gate 10).
4. **Recovery assumptions.** Repossession and resale rates cut to observed levels until a six month repossession programme demonstrates otherwise.
5. **Affordability redesign.** The Tier 2 to 4 price plans are restructured, through longer tenors or higher deposits, so that the instalment stays within 10% of target segment income, with incomes validated by customer surveys.
6. **Valuation protection.** A pre money valuation closer to USD 6m, or a ratchet tied to Year 2 cohort performance.
7. **Data and validation.** Monthly updates of *Credit_Input* and *Vintage_Input*, ownership tracking at twice the tenor from launch, independent review of the ECL approach (gate 19) and of the tax and accounting treatment (gate 15), a full test of the workbook in Excel (gate 17) and management sign off (gate 2).

Readiness today. Four of the 23 gates are met: model integrity, positive lifetime contribution in every tier, a credit engine running on actual data, and data reconciliation. Gate 10, no covenant breach in the active scenario, is not met: the monthly covenants hold in Base, but the annual DSCR is below the 1.20x minimum in Years 1 to 4. The workbook's own rule therefore reads STOP on a failed test, and 9 of its 13 critical gates still lack evidence. That STOP is a finding against the business plan as modelled under the assumed facility terms, not only a gap in the file. The recommendation remains the analyst's judgement of what the investment needs, and it is explicitly conditional: the DSCR covenant must be replaced or reset with the lender (condition 3), and disbursement waits until the critical gates the conditions address are evidenced. The committee should see both. The conditions above close the most material gaps.

Accounting. Revenue recognition and ECL in the workbook are simplified. This is an analytical modelling treatment and does not constitute a determination of the applicable accounting treatment under IFRS.

10. Questions for discussion

1. Reset the credit assumptions to the management plan (default hazard and collection rate on *Products*). Which outputs move most, and why does the investor IRR shift by about four points?
2. Set FX pass through to 100% in Base. What happens to affordability on *Consumer_Risk*, and what does that imply for defaults?
3. Cut the Tier 2 tenor from 24 to 18 months at the same cash price. How do the payment burden, the expected loss rate and the unit IRR respond?
4. Switch the RBF design to repayment linked (mode 2), then to ownership linked (mode 3). Why does mode 3 pay nothing, and what data would SolaraPay need to start collecting?
5. Extend borrowing base eligibility to 90 DPD on *Credit_Assumptions*. Why does Base barely change, and why would a lender still refuse the request?
6. Using the 24 month history, draft the collections covenant you would accept as a lender, and justify both the level and the cure period.

Appendix. Case files and data

File	Content
SolaraPay_Case_Model_v0.8-dev.xlsx	Case workbook on the calibrated inputs, with the synthetic history loaded in <i>Credit_Input</i> and <i>Vintage_Input</i>
case_exhibits.json	The figures quoted in this case, extracted from the evaluated workbook, together with the management plan comparison
figures/	The five charts reproduced in this document

Synthetic data. The history applies the calibrated credit behaviour with random variation under a fixed seed (20261002): 24 months of portfolio data for Tiers 1 to 3 and eighteen cohorts per tier observed at checkpoints from M3 to M24. It is teaching material only and describes no real company.

Management plan comparison. The management plan figures in Sections 2, 5 and 10 come from the same model run on management's original credit assumptions; only the default hazard and collection rates of Tiers 1 to 3 differ from the calibrated case.